

B.Tech. Degree IV Semester Examination April 2016

CS 1404 AUTOMATA LANGUAGES AND COMPUTATIONS (2012 Scheme)

Time : 3 Hours

Maximum Marks : 100

PART A

(Answer **ALL** questions)

(8 × 5 = 40)

- I. (a) Design a DFA to accept the language
 $L = \{w \mid w \text{ is of even length and begins with } 01\}$
- (b) Differentiate Moore and Mealy machine.
- (c) State whether the CFG: $A \rightarrow A + A \mid A - A \mid a$. is ambiguous and justify your answer.
- (d) Define Greibach normal form. Convert the following productions in CFG to GNF $S \rightarrow abSb \mid aa$
- (e) What is a universal turing machine?
- (f) Write down church's hypothesis.
- (g) What do you mean by Halting problem of a turing machine?
- (h) Give short notes on recursively enumerable language.

PART B

(4 × 15 = 60)

- II. (a) Prove the equivalence of NFA and DFA. (7)
- (b) Construct the DFA equivalent to the NFA $M = (\{q_0, q_1\}, \{a, b\}, \delta, q_0, \{q_1\})$ where (8)
- δ is given by

	a	b
q ₀	q ₀ , q ₁	q ₁
q ₁	q ₀	-

OR

- III. (a) State pumping lemma for regular expression. Given $L = \{a^n b^n \mid n \geq 1\}$ where (10)
- $\{a, b\}$ is the alphabet. Is L regular? If yes, draw the DFA for L. Otherwise, prove that L is not regular using pumping lemma.
- (b) Define regular grammar and show that class of language by FA is closed under (5)
- concatenation.
- IV. (a) Give CFG for generating the set of all strings over alphabet $\{a, b\}$ with exactly (5)
- twice as many a's as b's.
- (b) State and prove equivalence of acceptance by final state and empty stack in (10)
- PDA.

OR

(P.T.O.)

- V. (a) Construct a PDA equivalent to the following grammar $S \rightarrow aSbb \mid a$. (6)
 Design a PDA to accept the language $L = \{a^n b^{n+1}; n \geq 1\}$. (9)
 (b)
- VI. (a) Formally define turing machine. Explain how turing machine can be used as a computer of integer functions. (7)
 (b) Design a turing machine to recognize the language $\{a^n b^{n+1} \mid n \geq 1\}$. (8)
- OR**
- VII. Explain any three methods of construction of turing machine. (15)
- VIII. (a) Discuss Chomsky Hierarchy with typical examples. (8)
 (b) Explain how a regular grammar is converted to a finite automata with the help of an example (7)
- OR**
- IX. (a) State and prove equivalence of regular grammar and FA. (10)
 (b) Define linear bound automata. (5)

